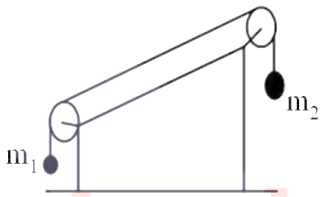
Sec: SR.IIT _*(MODEL-A) **Date: 06-01-26**
Time: 3 Hrs **SPL GTM TEST** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

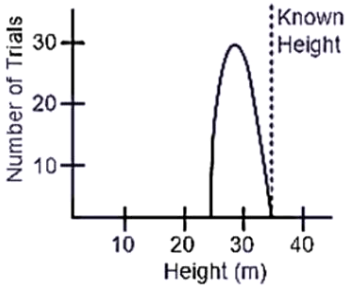
1. Two masses are attached with pulleys by a massless rope on an inclined plane as shown. All surfaces are frictionless. If the masses are released from rest, then the inclined plane.



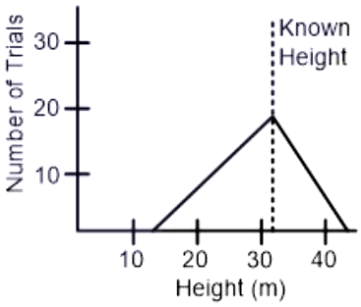
(A) Accelerates to the left if $m_1 < m_2$
(B) Accelerates to the right if $m_1 < m_2$
(C) Accelerates to the left regardless of the masses
(D) Does not move

2. Students of four classes measure the height of a building. Each class uses different method and each measures the height many times. The data for each class are plotted below. Which class made the most precise measurement?

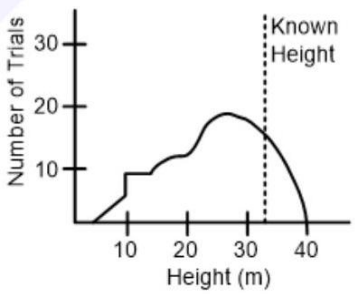
(A)



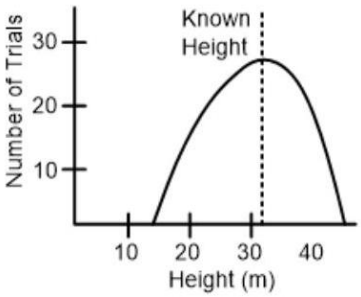
(B)



For More Join: @IITJEE_Advanced



(C)



(D)

3. **Assertion** : If volume of a gas is increasing but temperature of the gas is decreasing, then heat given to the gas may be positive, negative or zero.

Reason : Heat given to a gas is a state function. It is not path function.

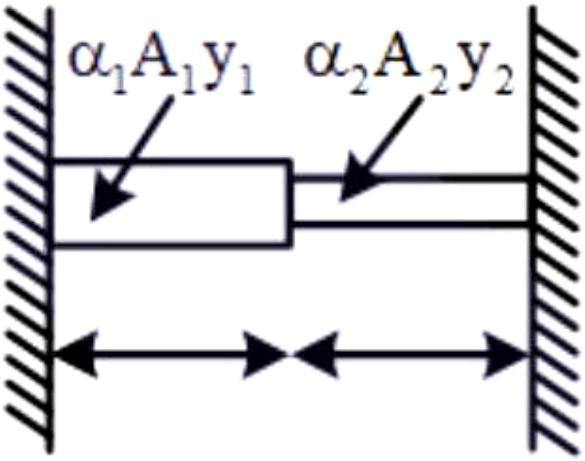
(A) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

(B) Assertion is incorrect but Reason is correct.

(C) Assertion is correct but Reason is incorrect.

(D) Both Assertion and Reason are correct but reason is not the correct explanation of Assertion.

4. Two elastic rods are joined between fixed supports as shown in figure. Condition for no change in lengths of individual roads with the increase of temperature is :($\alpha_1, \alpha_2 =$ linear expansion coefficient, $A_1, A_2 =$ area of rods, $Y_1, Y_2 =$ Youngs modulus,)



(A) $\frac{A_1}{A_2} = \frac{\alpha_1}{\alpha_2}\frac{y_1}{y_2}$

(B) $\frac{A_1}{A_2} = \frac{L_1\alpha_1 y_1}{L_2\alpha_2 y_2}$

(C) $\frac{A_1}{A_2} = \frac{L_2\alpha_2 y_2}{L_1\alpha_1 y_1}$

(D) $\frac{A_1}{A_2} = \frac{\alpha_2 y_2}{\alpha_1 y_1}$

For More Join: @IITJEE_Advanced

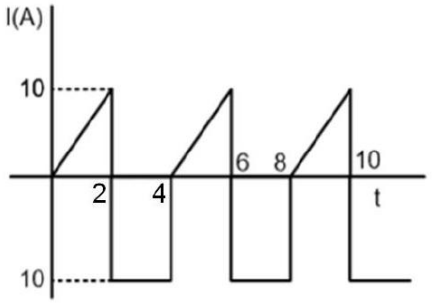
5. In YDSE width of one slit is four times width of 2nd slit then fringe visibility $= \frac{I_{max} - I_{min}}{I_{max} + I_{min}}$ will be.

(A) 60% (B) 80% (C) 50% (D) 100%

6. In a new unit system 1 unit of time is equal to 10 second, 1 unit of mass is 5 kg and 1 unit of length is 20 m . In this new system of units, 1 unit of energy is equal to

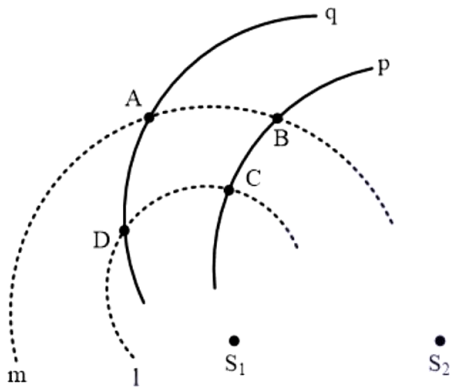
(A) 20 Joule (B) $\frac{1}{20}$ Joule (C) 4 Joule (D) 16 Joule

7. The value of I_{rms} is



(A) 8.16 A (B) 16.32 A (C) 4 A (D) 8 A

8. In the figure shown, S_1 and S_2 are two identical coherent point sources of light which are coherent and out of phase by 180°. Taking S_1 as centre, two circular arcs ℓ and m of radii 1 m and 2 m are drawn. Taking S_2 as centre, two circular arcs p and q are drawn having radii 2 m and 4 m respectively. It is given that wavelength of wave produced by each source is 4.0 m . Then Choose correct options among the following.



(A)Constructive interference is observed at A,

(B) Constructive interference is observed at B

For More Join: @IITJEE_Advanced

(C) Constructive interference is observed at C

(D) Constructive interference is observed at D

9. A planet of mass M , has two natural satellites with masses m_1 and m_2. The radii of their circular orbits are R_1 and R_2 respectively. Ignore the gravitational force between the satellites. Define v_1, L_1, K_1 and T_1 to be, respectively, the orbital speed, angular momentum, kinetic energy and time period of revolution of satellite 1 ; and v_2, L_2, K_2 and T_2 to be the corresponding quantities of satellite 2. Given $m_1/m_2 = 2$ and $R_1/R_2 = 1/4$, match the ratios in List-I to the numbers in List-II.

	List – I		List – II
P	$\frac{v_1}{v_2}$	1	$\frac{1}{8}$
Q	$\frac{L_1}{L_2}$	2	1
R	$\frac{K_1}{K_2}$	3	2
S	$\frac{T_1}{T_2}$	4	8

(A) $P \to 4; Q \to 2; R \to 1; S \to 3$ (B) $P \to 3; Q \to 2; R \to 4; S \to 1$

(C) $P \to 3; Q \to 3; R \to 1; S \to 4$ (D) $P \to 3; Q \to 3; R \to 4; S \to 1$

10. The electric field due to the electric potential $V = 2x^2 - 4x$ is

(A) $(4x+4)\hat{i}$ (B) $(4x-4)\hat{i}$ (C) $(-4x+4)\hat{i}$ (D) $(-4x-4)\hat{i}$

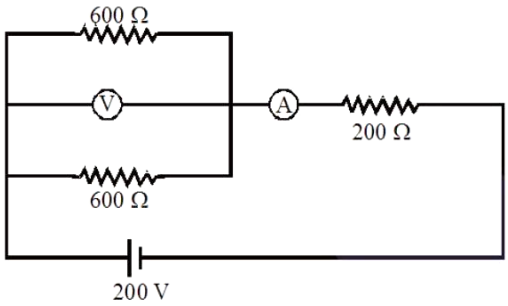
11. A charge of $1\mu C$ is placed at point $P(3,2,5)$. Find the electric field vector at point $S(0,-4,3)$:-

(A) $\frac{-9}{343}\left(3\hat{i}+6\hat{j}+2\hat{k}\right)\times 10^3$ N/c (B) $\frac{9}{343}\left(3\hat{i}-6\hat{j}+\hat{k}\right)\times 10^3$ N/c

(C) $\frac{-3}{343}\left(3\hat{i}+6\hat{j}+2\hat{k}\right)\times 10^3$ N/c (D) $\frac{9}{343}\left(3\hat{i}+6\hat{j}+2\hat{k}\right)\times 10^3$ N/c

For More Join: @IITJEE_Advanced

12. Assuming the cell and ammeter to be ideal and resistance of voltmeter is 600Ω, the reading of voltmeter and ammeter are respectively :-

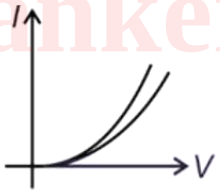
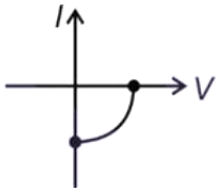
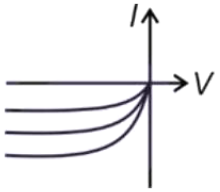
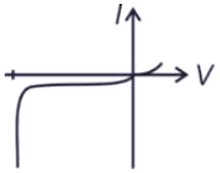


(A) 100 V, 0.5 A (B) 200 V, 0 (C) 200 V, 0.5 A (D) 0, 0

13. Energy required for electron excitation in Li^{++} from the first to the third Bohr orbit is :-

(A) 36.3 eV (B) 108.8 eV (C) 122.4 eV (D) 10.8 eV

14. Some V-I characteristics of special purpose diodes are given in List-I and the name of the diodes are given in List-II. Match the List-I with List-II.

	List-I		List-II
A.		P.	Zener diode
B.		Q.	Solar cell
C.		R.	LED
D.		S.	Photodiode

(A) $A \to R, B \to P, C \to Q, D \to S$ (B) $A \to S, B \to R, C \to Q, D \to P$

(C) $A \to P, B \to Q, C \to R, D \to S$ (D) $A \to R, B \to Q, C \to S, D \to P$

For More Join: @IITJEE_Advanced

15. Which of these are correct?
I. In forward biasing holes from p-side crosses junction and reach n-side.
II. In forward biasing electron from n -side crosses junction and reach p-side.
III. In n -side holes are minority carriers.
IV. In p -side electrons are minority carriers.
(A) I, II and III (B) I, III and IV
(C) II, III and IV (D) I, II, III and IV

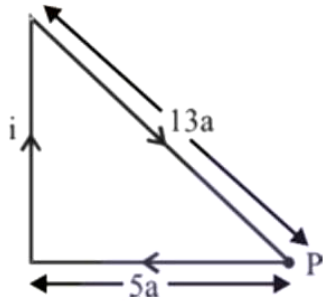
16. When an oscillator completes 100 oscillation its amplitude reduced to $\frac{1}{3}$ of initial value. What will be its amplitude, when it completes 200 oscillation?
(A) $\frac{1}{8}$ (B) $\frac{2}{3}$ (C) $\frac{1}{6}$ (D) $\frac{1}{9}$

17. The electric field in an electromagnetic wave is given by $E = 56.5\sin\omega(t - x/c)\,NC^{-1}$. Find the intensity of the wave if it is propagating along x -axis in the free space.
(A) $5.65 Wm^{-2}$ (B) $4.24 Wm^{-2}$ (C) $1.9\times10^{-7} Wm^{-2}$ (D) $56.5 Wm^{-2}$

18. A lensmaker has to design an achromatic combination of effective power -0.8 D . The two types of glasses to be used (A and B) have dispersive powers in the ratio $2:3$. Then the lens made from glass B will have focal length :-
(A) 41.6 cm (B) 62.5 cm (C) -41.6 cm (D) -62.5 cm

19. When a ferromagnetic material is heated to temperature above Curie temperature, the material :-
(A) Is permanently magnetized
(B) Remains ferromagnetic material
(C) Bahaves like a diamagnetic material
(D) Behave like a paramagnetic material

20. Find magnetic field at point P .



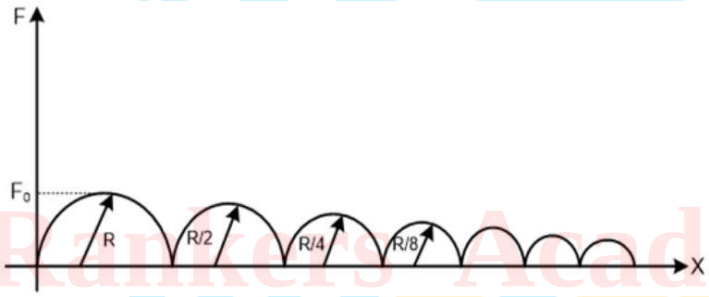
(A) $\frac{3\mu_0 i}{65\pi a}$ (B) $\frac{\mu_0 i}{39\pi a}$ (C) $\frac{4\mu_0 i}{65\pi a}$ (D) $\frac{12\mu_0 i}{65\pi a}$

For More Join: @IITJEE_Advanced

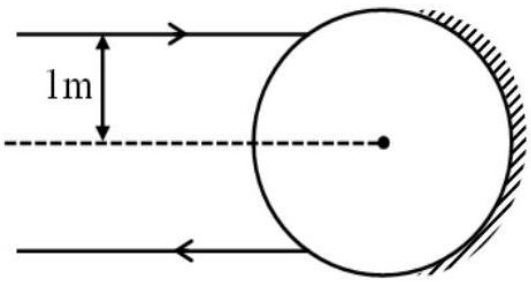
SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. A particle of mass m_0 is moving in a straight line under the influence of force F. Its force (F) - displacement (x) variation is shown in the diagram, which is in the form of semicircles with centre on x -axis and having radius $\left[R, \frac{R}{2}, \frac{R}{4}, \frac{R}{8} \ldots\ldots\ldots\ldots \infty\right]$. Particle starts form $x = 0$ with $u = 0^+$. If speed of the particle at ∞ is given by $v = \sqrt{\frac{A\pi F_0^2}{Bm_0}}$, then the value of $(A + B)$ is :



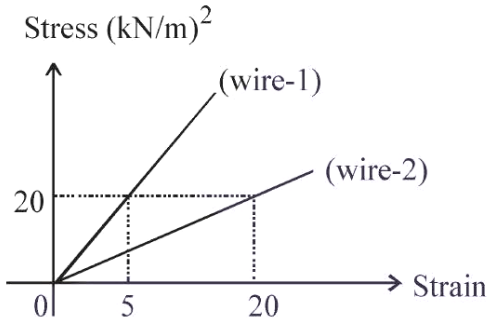
22. A transparent cylinder of radius $R = 2.00\ m$ has a mirrored surface on its right half as shown in the figure. A light ray traveling in the air is incident on the left side of the cylinder. The incident ray and the exiting ray are parallel and at a distance $d = 2.00\ m$. The refractive index of the material is $\mu = x\cos 15^\circ$. Find x .



23. Moment of inertia of a thin uniform hexagonal plate of side $a = \frac{2}{\sqrt{5}}\ m$ & mass $M = 12\ kg$ about an axis passing through the centre & perpendicular to the plane is (In kgm^2)

For More Join: @IITJEE_Advanced

24. For two metal wires, stress-strain curve is shown in figure.



Find ratio of extension in wire-1 to wire-2, when stress of 40kN/m^2 and 10kN/m^2 are applied on wire-1 and wire-2 respectively. Ratio of length of wire-1 and wire-2 is $2:1$.

25. Equation of trajectory of a projectile thrown from ground is given as $2x^2-8x+3y=0$ (x and y in metre). If point of projection is origin, range of projectile (in m) will be :-

CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Which of the following statement is correct?

(A) Electron gain enthalphy of oxygen is more negative than that of sulphur.

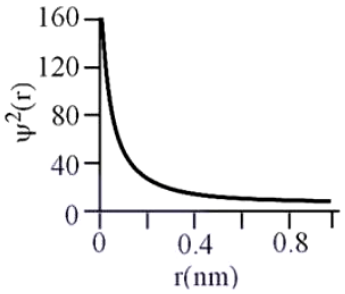
(B) Electronegativity of fluorine is less than chlorine.

(C) Ionisation enthalpy of nitrogen is less than oxygen.

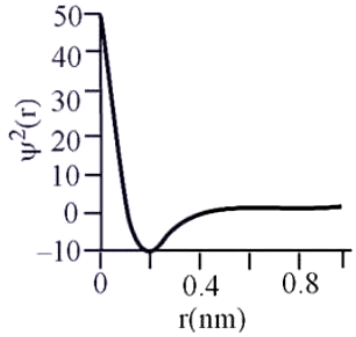
(D) Atomic size of $Na^+(g)$ is smaller than $Na(g)$

27. Which of the following graphs correctly represents the variation of electron probability density with the distance from the nucleus for a 2 s orbital?

(A)

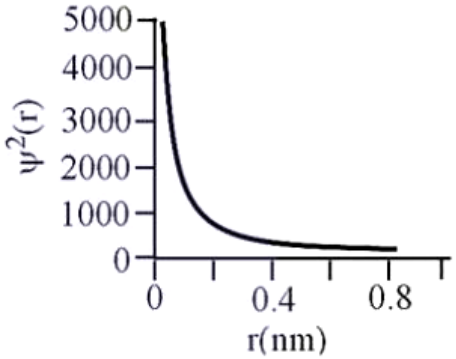


(B)

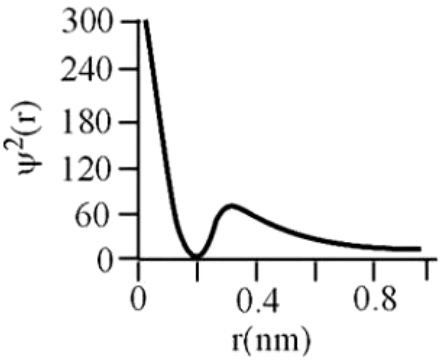


For More Join: @IITJEE_Advanced

(C)



(D)



28. Given below are two statements; one is labelled as

Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Stability of +5 oxidation state decreases and that of +3 oxidation state increase down the group in group 15.

Reason (R) : Inert pair effect is more prone in case of heavier members of p -block.

In the light of the above statements, choose the most appropriate answer from the options given below:

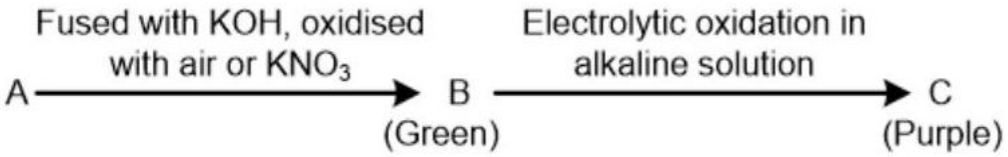
(A) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(B) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(C) (A) is correct but (R) is not correct.

(D) (A) is not correct but (R) is correct.

29. Find the compound ' A ' from the following reaction sequences.



(A) MnO (B) MnO_2 (C) CrO_3 (D) MnS

30. The Conductivity of 0.01 M HA is 1×10^{-5} S cm^{-1}. If Λ_m^0 for HA is 100 S cm^2 mol^{-1}. Then % dissociation of HA is

(A) 5% (B) 2% (C) 1% (D) 10%

For More Join: @IITJEE_Advanced

31. Correct statement is

(A) Ratio of molar concentrations of dissociated water to undissociated water is 1.8×10^{-5} at 298 K

(B) pK_a of CH_3COOH is 4.74 , then pK_b of $CH_3COO^{\ominus}$ is 9.26 at 298 K

(C) $CH_4 < NH_3 < HF < H_2O$ (Acidic strength)

(D) Solubility of AgCl in pure water is less than that in 0.1 M NaCl solution

32. Which of the following reaction is having correct major product mentioned?

(i) $2R-OH + 2Na \rightarrow 2R-O-Na + H_2$

(ii) $CH_3CH=CH_2 \xrightarrow{H_2O/H^+} CH_3\underset{\displaystyle OH}{\underset{|}{C}}H-CH_3$

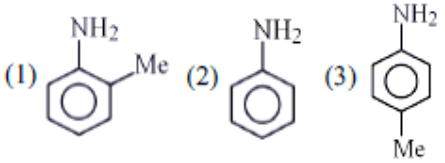
(iii) $CH_3CH=CH_2 \xrightarrow[H_2O_2/OH^-]{B_2H_6} \underset{\text{(Major)}}{CH_3CH_2CH_2OH}$

(iv) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{C}}-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3 \xrightarrow[\Delta]{H^+} \underset{\text{(Major)}}{CH_3-\underset{\displaystyle CH_3}{\underset{|}{C}}-CH=CH_2}$

(A) (i), (ii) only (B) (i), (ii) (iii) only

(C) (ii), (iv) only (D) (i) and (iii) only

33. Compare basic strength order of



(A) 3 > 1 > 2 (B) 3 > 2 > 1 (C) 2 > 1 > 3 (D) 1 > 3 > 2

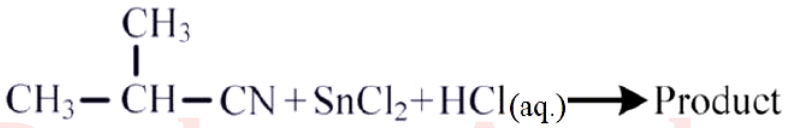
For More Join: @IITJEE_Advanced

34. Match the column

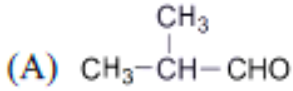
Column-I (Carbohydrates)		Column-II (Source or name)	
(a)	Sucrose	(p)	Animal starch
(b)	Lactose	(q)	Cereals, roots, tubers
(c)	Glycogen	(r)	Milk sugar
(d)	Starch	(s)	Cane sugar

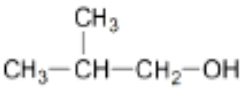
(A) $a-r, b-s, c-q, d-p$ (B) $a-s, b-r, c-p, d-q$

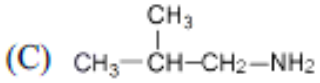
(C) $a-s, b-p, c-r, d-q$ (D) $a-p, b-q, c-r, d-s$

35. 

product is

(A) 

(B) 

(C) 

(D) None of these

36. Which of the following statement is correct?

(A) $Li < Be < B < C\,(IE_1)$

(B) Electron gain enthalpy of Fluorine is more negative than chlorine.

(C) All the Alkaline earth metal hydroxides are insoluble in NaOH

(D) Al_2O_3 and As_2O_3 both are amphoteric

For More Join: @IITJEE_Advanced

37. Which of the following Molecule have Bond order two as well as both the bonds are Pi (π) bonds?

(A) O_2 (B) B_2 (C) C_2 (D) N_2

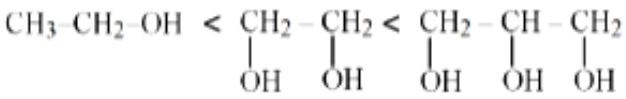
38. Anion ' X ' on treating with concentrated H_2SO_4 gives deep violet vapour which turn starch paper blue. During the reaction of anion ' X ' with H_2SO_4, some other products are also formed. These products are

(A) S, H_2S, IO_3^- (B) H_2S, SO_3

(C) $HI, H_2S, S_2O_3^{2-}$ (D) S, H_2S, HI, SO_2

39. (I) Lower aldehyde are more reactive than higher aldehyde for nucleophilic addition reaction

(II) Order of boiling point



(III) Methanol is more poisonous than ethanol.

(IV) As molar mass of 1° alcohol increases its solubility decreases.

Correct statement from the above are :

(A) I, II (B) I, III, IV (C) II, III (D) II, IV

40. Statement-I: One should read lower meniscus in case of colourless and transparent solution and upper meniscus for solutions of dark colors.

Statement-II: A secondary standard can be used for preparing standard solution by direct weighing

(A) Both statement-I and statement-II are correct.

(B) Both statement-I and statement-II are incorrect.

(C) Statement-I is incorrect and statement-II is correct.

(D) Statement-I is correct and statement-II is incorrect.

41. In paper chromatography, the stationary phase and mobile phase are respectively

(A) finley divided solid, liquid (B) liquid, solid

(C) liquid, liquid (D) solid, solid

For More Join: @IITJEE_Advanced

42. Pick the incorrect statement from following.

(A) C_2, Li_2, O_2^{2+} are diamagnetic

(B) $MgO < Al_2O_3 < SiO_2$ (Acidic nature)

(C) Shape of IF_7 is pentagonal bipyrimidal

(D) Hybridisation of Xe in cationic part of solid XeF_6 is sp^3d.

43. Which of the following theory explains the bonding in metal carbonyl most appropriately?

(A) Valence bond theory (B) Crystal field theory

(C) Molecular orbital theory (D) Ligand field theory

44. The average degree of freedom per molecule for a gas is 6 . The gas performs 30 Joule of work when it expands at constant pressure. What is the amount of heat absorbed by the gas?

(A) 120 J (B) -120 J (C) 90 J (D) -90 J

45. The equilibrium constants for the reaction $A_3 \rightleftharpoons 3A$ at 300 K and 700 K are 2×10^{-10} and 1×10^{-5}, respectively. The reaction must be :

(A) Very fast (B) Endothermic (C) Very slow (D) Exothermic

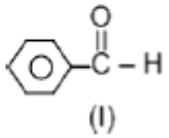
SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

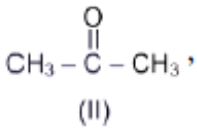
This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

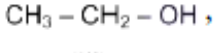
46. A catalyst lowers the activation energy by 10 kJ / mole . Enthalpy of reaction $(\Delta H)\, A \rightleftharpoons B$ is +25 kJ / mole without catalyst. Final Enthalpy of reaction $(\Delta H)\, A \rightleftharpoons B$ after the addition of catalyst. (in kJ / mole) is

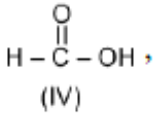
For More Join: @IITJEE_Advanced

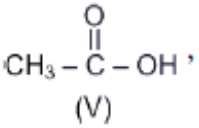
47. From the following compounds

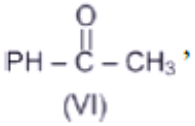

(I)

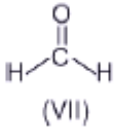

(II)


(III)


(IV)


(V)


(VI)

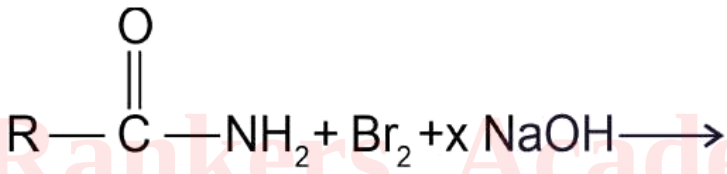

(VII)

a = Total no. of compounds show positive tollens test

b = Total no. of compounds show positive haloform test

Find $a+b$

48. The following reaction takes place during Hoffmann bromamide reaction.



$$RNH_2 + Na_2CO_3 + yNaBr + zH_2O$$

Find the ratio of x to y.

49. The rate constant of a first order reaction is $2\times10^{-3}\ s^{-1}$ and $0.24\ min^{-1}$ at $27^{\circ}C$ and $57^{\circ}C$, respectively. The activation energy of reaction (in $kcal/mol$) is ____.

($\ln 2 = 0.70$)

(*Answer to the nearest integer*)

50. Liquids A and B form an ideal solution. At $40^{\circ}C$, the mole-fractions of A in liquid and vapour forms is 0.4 and 0.6 , respectively. If the total vapour pressure of solution at $40^{\circ}C$ is 500 torr, then the vapour pressure of pure liquid B at $40^{\circ}C$ is ____ torr.

(Answer to the nearest integer)

Rankers Academy JEE
THE NARAYANA GROUP

For More Join: @IITJEE_Advanced

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Let $A(2,3)$ and $B(6,1)$ be two vertices of a ΔABC lying in the 1st quadrant with $\angle ABC = 90^\circ$, the area of $\Delta ABC = 10\sqrt{2}$ sq. units. If vertex C lies in the 1st quadrant, then the ordinate of point ' C ' is

(A) $3+2\sqrt{2}$ (B) $2+3\sqrt{2}$ (C) $5+\sqrt{2}$ (D) $1+4\sqrt{2}$

52. Consider three vectors $\vec{a} = \hat{i}+2\hat{j}+\hat{k}; \vec{b} = 2\hat{i}-\hat{j}+\hat{k}$ and $\vec{c} = \hat{i}+\hat{j}+\hat{k}$. Let $\vec{x}$ be a vector lying in plane of $\vec{a}$ and $\vec{b}$ and perpendicular to the vector $\vec{c}$, then length of projection of a unit vector along $\vec{x}$ on the $\vec{a}$ is

(A) $\frac{2}{\sqrt{26}}$ (B) $\frac{2}{\sqrt{39}}$ (C) $\frac{3}{\sqrt{39}}$ (D) None of these

53. Assertion (A) : Number of ways of selection of any 4 points from 10 collinear points such that no two of them are consecutive is 35

Reason (R) : Number of ways of choosing k number of points from n collinear points such that no two of them are consecutive is ${}^{n-k+1}C_k$

(A) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(B) Both Assertion and Reason are true and Reason is not the correct explanation of Assertion.

(C) Assertion is true but Reason is false

(D) Assertion and Reason both are false.

54. If $f(x) = (2x-3\pi)^5 + \frac{4}{3}x + \cos x$ and g is the inverse function of f, then $g'(2\pi)$ is equal to

(A) $\frac{7}{3}$ (B) $\frac{3}{7}$ (C) $\frac{30\pi^4+4}{3}$ (D) $\frac{3}{30\pi^4+4}$

For More Join: @IITJEE_Advanced

55. If $f(x)=\int_0^1 e^{|t-x|}dt$ where $0\le x\le 1$, then maximum value of $f(x)$ is

(A) e^{-2} (B) e^{-3} (C) $e-1$ (D) $2\left(\sqrt{e}-1\right)$

56. Let z_1,z_2,z_3 be roots of the equation $z^3-3z+1=0$, then value of $\sum_{k=1}^{3}\frac{z_k}{1-z_k}$ equals

(A) -1 (B) 0 (C) -3 (D) 1

57. Let A_n be the area of region bounded by a curve $y=x^3\left(1-x^2\right)^n, 0\leqslant x\leqslant 1$ and the x -axis, then the value of $\sum_{n=1}^{\infty}A_n$ is equal to

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) 1

58. Let $f(x)=x^2+xg'(1)+g''(2)$ and $g(x)=x^2f(1)+xf'(x)+f''(x)$, then $\left(\frac{1-f''(3)}{1-g''(2)}\right)$ equals

(A) -1 (B) -2 (C) 4 (D) -3

59. Consider the data set $\{8,9,10,11,12\}$

	List – I		List – II
(A)	If we double each observation of given data set then new mean of newly formed data set	(I)	20
(B)	If we replace 8 and 12 by mean of given data set then new mean	(II)	2
(C)	Variance of given data set is	(III)	10
(D)	Standard deviation of data set whose entries are double of the entries of given data set	(IV)	$2\sqrt{2}$

(A) A – I; B – III; C – II; D – IV (B) A – I; B – II; C – III; D – IV

(C) A – II; B – III; C – I; D – IV (D) A – III; B – II; C – IV; D – I

For More Join: @IITJEE_Advanced

60. If α,β are roots of the equation

$$x^2-\left(6-\frac{1}{7^{\sqrt{\log_7 5}}-5^{\sqrt{\log_5 7}}+1}\right)x+\frac{1}{5^{(\log_5 7)^{\frac{1}{3}}}-7^{(\log_7 5)^{\frac{2}{3}}}-\frac{1}{3}}=0,$$

then the equation whose roots are $\alpha+\frac{1}{\beta}$ and $\beta+\frac{1}{\alpha}$ is

(A) $3x^2-20x-12=0$ (B) $3x^2-10x-4=0$

(C) $3x^2-10x+2=0$ (D) None of these

61. A normal of slope m is drawn to $y^2=4ax$ at point P , cut x -axis at point Q . If area of triangle PQS where S is the focus of the parabola $y^2=4ax$ is A , then (a, m, A) cannot be

(A) $(3,2,90)$ (B) $(2,3,120)$ (C) $(2,2,40)$ (D) $(3,4,610)$

62. Let ' P ' be the image of origin $(0,0,0)$ w.r.t. the line $\frac{x-1}{1}=\frac{y-2}{1}=\frac{z-3}{-1}$, then distance of point P from origin equals

(A) $\sqrt{11}$ (B) $\sqrt{13}$ (C) $2\sqrt{14}$ (D) $\sqrt{14}$

63. The value of $\sum_{i=0}^{p} {}^{m}C_i\,{}^{n+i}C_p\,{}^{p+n}C_{p-i}$ is equal to

(A) ${}^{m+p}C_n\,{}^{n+p}C_p$ (B) ${}^{n+p}C_n\,{}^{m+n}C_p$ (C) ${}^{n+p}C_m\,{}^{m+n}C_n$ (D) ${}^{m+p}C_m\,{}^{m+n}C_n$

64. If $\tan^{-1}x^2+\tan^{-1}y^2=[xy]\frac{\pi}{4}$, then sum of maximum and minimum integral value of $\left(x^2+y^2\right)$ is (where $x,y>0$ and [.] denotes greatest integer function)

(A) 23 (B) 22 (C) 24 (D) 21

65. Given below are two statements :

Statement-I : In a triangle ABC , let D be mid-point of side BC then it follows $\left(AB^2+AC^2\right)=\left(2AD^2+\frac{1}{2}BC^2\right)$.

Statement-II : Image of orthocentre in any side of the triangle will always lies on the circumcircle of the triangle.

Choose the most appropriate answer from the options.

(A) Both statement-I and statement-II are correct.

(B) Both statement-I and statement-II are incorrect.

For More Join: @IITJEE_Advanced

(C) Statement-I is incorrect and statement-II is correct.

(D) Statement-I is correct and statement-II is incorrect.

66. If $\lim\limits_{x\to 0}\left[\frac{10\sin^{-1}x}{x}\right]+\left[\frac{8\sin^{-1}2x}{x}\right]+\left[\frac{12x}{\tan^{-1}2x}\right]=\ell$, then ℓ is equal to

(where [.] denotes greatest integer function)

(A) 30 (B) 31 (C) 32 (D) 33

67. The integral $\frac{\int_0^{\log\left(1+\sqrt{2}\right)}\left(e^t+e^{-t}\right)^8\,dt}{\int_0^{\pi/4}\sec^9 x\,dx}$ simplifies to

(A) 256 (B) 128 (C) 512 (D) 1024

68. If the first 3 consecutive terms of a geometrical progression are the roots of equation $2x^3-19x^2+57x-54=0$, then sum of the infinite G.P. is

(A) 9 (B) $\frac{27}{2}$ (C) $\frac{9}{2}$ (D) 27

69. Solution of differential equation $xdx+ydy=xdy-ydx$ which passes through the point $(1,0)$ is -

(A) $x^2+y^2=\frac{2}{3}x^2y+1$ (B) $\frac{1}{x^2+y^2}=2\tan^{-1}\frac{y}{x}+1$

(C) $\ln\left(x^2+y^2\right)=2\tan^{-1}\frac{y}{x}$ (D) $x^2+y^2=2xy+1$

70. If matrix A commutes with matrix $B=\begin{bmatrix}1&0&0\\1&1&0\\1&1&1\end{bmatrix}$ such that $|A|=8$, then $\mathrm{tr}(A)$ is (where $\mathrm{tr}(A)$ denotes trace of matrix A)

(A) 6 (B) 5 (C) 7 (D) 4

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:

Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

For More Join: @IITJEE_Advanced

71. A bag contains 15 unbiased coins and 2 coins with head on both sides, and 3 coins with tail on both sides. One coin is selected at random and tossed once, shows HEAD . If the probability that the selected coin was unbiased is $\frac{m}{n}$, where $\gcd(m,n)=1$, then the value of ($n-m$) equals

72. Let $\int(\cot 2x\cdot\cot 3x-\tan 2x\cdot\tan 7x)\tan 5x dx=a\ell n(\tan 2x)+b\ell n(\sin 3x)+c\ell n(\sec 5x)+d\ell n(\cos 7x)$ +C and $a,b,c,d\in Q$ and C is constant of integration. If $(a+b+c+d)$ can be expressed as $\frac{m}{n}$ in the simplest form, then $(m+n)$ equals (where $m,n\in N$)

73. The number of real values of x for which $2^{(2^x-1)^2}+4^x=\sqrt{x}+2^{x+1}+\log_2(1+\sqrt{x})$ is

74. $f:[-2,2]\to R, f(x)=[x]^2+|x|$

If $m=$ number of integer in the range of $f(x)$

n=number of points at which the function is discontinuous (where [.] represents greatest integer function)

then ($m+n$) equals

75. Number of solution(s) of the equation $f(f(x))=0$ if $f(x)=\frac{x^3}{3}-\frac{x^2}{2}+1$ is

SRAMAYEVA JAYATHE
Rankers Academy JEE
THE NARAYANA GROUP

For More Join: @IITJEE_Advanced